Matplotlib basics

Choose the plot type that matches your question.

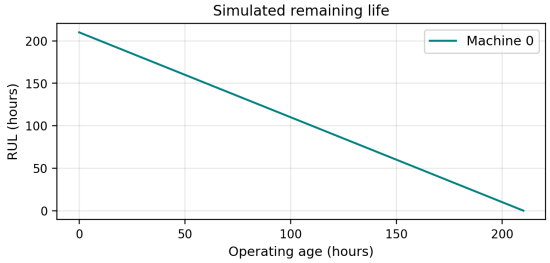
Open ML4EngApp in VS Code. Keep the CSV beside your .ipynb notebook and choose the MachineLearning4EngApp kernel. Run each block as a Code cell with Shift+Enter; # marks a comment.

```
import pandas as pd  # read and manipulate tables
import numpy as np   # numeric calculations
import matplotlib.pyplot as plt  # plotting tools
df = pd.read_csv("vibration_rul_dataset.csv")
m0 = df.loc[df["machine_id"] == 0].sort_values("age_hours")
```

fig is the whole figure; ax is one plotting area. figsize is width and height in inches. ax.set(...) adds labels and a title. These outputs use the supplied synthetic vibration CSV.

1 Line plot

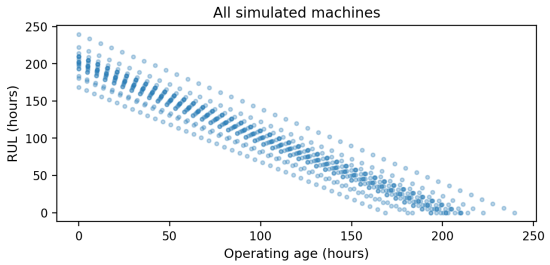
```
fig, ax = plt.subplots(figsize=(6, 3))  # figure + Axes
ax.plot(m0["age_hours"], m0["rul_hours"],
        color="teal", label="Machine 0")  # join points
ax.set(xlabel="Operating age (hours)", ylabel="RUL (hours)",
       title="Simulated remaining life")  # label the plot
ax.grid(True, alpha=0.3)  # faint reference lines
ax.legend()  # show the label supplied to plot()
fig.tight_layout(); plt.show()  # fit labels; display
```



Connect ordered observations; use a legend when lines have labels.

2 Scatter plot

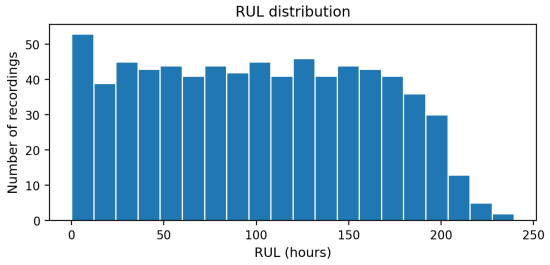
```
fig, ax = plt.subplots(figsize=(6, 3))
ax.scatter(df["age_hours"], df["rul_hours"],
           s=8, alpha=0.3)  # marker size and opacity
ax.set(xlabel="Operating age (hours)", ylabel="RUL (hours)",
       title="All simulated machines")
fig.tight_layout(); plt.show()
```



Show relationships without joining points. alpha=0.3 makes overlaps visible.

3 Histogram

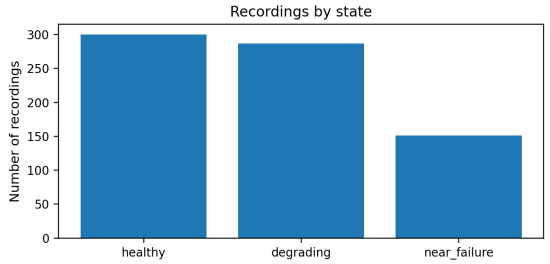
```
fig, ax = plt.subplots(figsize=(6, 3))
ax.hist(df["rul_hours"], bins=20,
        edgecolor="white")  # count values in 20 intervals
ax.set(xlabel="RUL (hours)", ylabel="Number of recordings",
       title="RUL distribution")
fig.tight_layout(); plt.show()
```



Count values in numeric intervals. Try bins=10 or bins=40.

4 Bar chart

```
counts = df["state"].value_counts()  # count each category
fig, ax = plt.subplots(figsize=(6, 3))
ax.bar(counts.index, counts.values)  # names and heights
ax.set(ylabel="Number of recordings",
       title="Recordings by state")
fig.tight_layout(); plt.show()
```



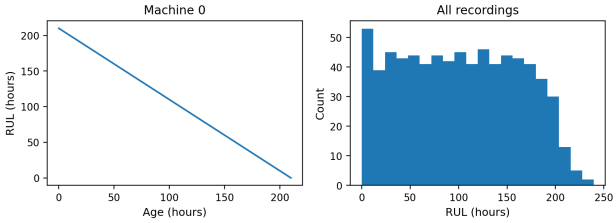
Compare counts across categories, rather than numeric intervals.

Compare plots and spot outliers

Subplots, boxplots, reproducibility and saving figures.

5 Two subplots in one figure

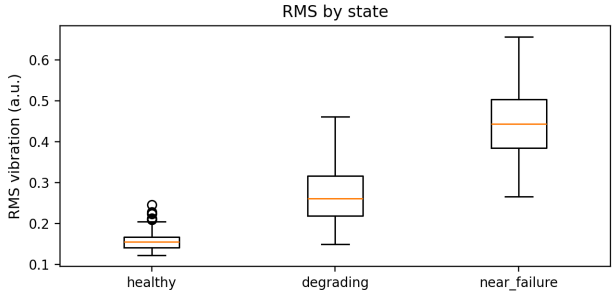
```
fig, axes = plt.subplots(1, 2, figsize=(8, 3))
axes[0].plot(m0["age_hours"], m0["rul_hours"])  # left plot
axes[0].set(title="Machine 0", xlabel="Age (hours)",
            ylabel="RUL (hours)")
axes[1].hist(df["rul_hours"], bins=20)  # right plot
axes[1].set(title="All recordings", xlabel="RUL (hours)",
            ylabel="Count")
fig.tight_layout(); plt.show()
```



axes[0] is the left plot; axes[1] is the right. Give each Axes its own labels. tight_layout() adjusts spacing so titles and tick labels fit.

6 Boxplots of vibration amplitude

```
# RMS: one vibration amplitude summary per recording
samples = [f"sample_{i:04d}" for i in range(1024)]
df["rms"] = np.sqrt((df[samples] ** 2).mean(axis=1))
states = ["healthy", "degrading", "near_failure"]
groups = [df.loc[df["state"] == s, "rms"] for s in states]
fig, ax = plt.subplots(figsize=(6, 3))
ax.boxplot(groups, whis=1.5, showfliers=True)
ax.set_xticks([1, 2, 3], states)  # label the three boxes
ax.set(ylabel="RMS vibration (a.u.)", title="RMS by state")
fig.tight_layout(); plt.show()
```



The sample list selects only sample_0000 to sample_1023. ** 2 squares values; mean(axis=1) averages across each row; sqrt gives its root mean square (RMS). Each group supplies one box.

Read the boxes: the middle line is the median; the box spans Q1 to Q3 (the middle 50%). IQR = Q3 - Q1. Whiskers reach the most extreme observations within 1.5 IQR of the box; dots beyond them are potential outliers.

7 Find and inspect potential outliers

```
healthy = df.loc[df["state"] == "healthy", "rms"]
q1, q3 = healthy.quantile([0.25, 0.75])  # quartiles
iqr = q3 - q1  # spread of the middle 50%
flagged = healthy.loc[(healthy < q1 - 1.5 * iqr) |
                      (healthy > q3 + 1.5 * iqr)]
print(len(flagged), "potential outlier recordings")
```

This dataset flags 8 healthy-state recordings. Inspect df.loc[flagged.index] to see them. A flagged point is not automatically an error: machine differences or degradation can explain it. Compare like-for-like groups and investigate before removing values.

8 Save figures and verify your notebook

Before plt.show(), add fig.savefig("my_plot.png", dpi=200, bbox_inches="tight"). dpi sets PNG resolution; bbox_inches trims whitespace. Use .pdf for vector output. Save your notebook with Ctrl+S.

Quick checks: x and y must have matching lengths. If a library is missing, run %pip install pandas numpy matplotlib, restart the kernel and Run All. Try machine 1, different histogram bins and another boxplot grouping.